



May 2024

(ME)

Max. Marks: 100

List of constants: Velocity of light, $c = 3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$, Electron mass, $m = 9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e = 1.6 \times 10^{-19} \text{ C}$, Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J/K}$, Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$, Permeability of the free space, $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$.

20 Marks

- $F = -kx$ is the force on a particle of mass 'm' undergoing SHM. What is the relationship between x and m in terms of angular frequency?**

A) $k = m\omega^2$ B) $k = m\sqrt{\omega}$
C) $m = k/\omega^2$ D) $m = k^2/\omega$
- Oscillations become damped due to**

A) Normal force B) Frictional force
C) Tangential force D) Parallel force
- A mass-spring system oscillates with a period of 2 seconds. What is the frequency of oscillation?**

A) 0.5 Hz B) 0.25 Hz
C) 1 Hz D) 2 Hz
- Number of oscillations per unit time is**

A) Amplitude B) Wavelength
C) Frequency D) Period
- The property by which a body returns to its original shape after removal of the force is called**

A) Plasticity B) Elasticity
C) Ductility D) Malleability
- The unit of Strain is**

A) Kg/cm^2 B) Newton/metre^2
C) Metre D) No unit.
- Which is correct Relation between Y (Young's Modulus), K (bulk modulus), n (rigidity modulus)**

A) $Y = \frac{9nK}{3K + n}$ B) $Y = \frac{9nK}{3K - n}$
C) $Y = \frac{9n + K}{3K - n}$ D) $Y = \frac{9n + K}{3K + n}$
- Twisting couple per unit twist of the cylinder is**

A) $C = \frac{2n \pi r^4}{L}$ B) $C = \frac{n \pi}{2L}$
C) $C = \frac{n \pi r^4}{2}$ D) $C = \frac{n \pi r^4}{2L}$
- An elemental semiconductor is formed by bonds.**

A) Covalent B) Electrovalent
C) Co-ordinate D) Ionic band

10. Addition of pentavalent impurity to a semiconductor creates many
 A) Valence electrons B) Holes
 C) Free electrons D) Bound electrons
11. When a pure semiconductor is heated, its resistance
 A) Goes up B) Goes down
 C) Remains the same D) Changes depending on atmospheric conditions
12. The mobility of electrons in a material is expressed in unit of:
 A) V/s B) m^2/s
 C) $\text{m}^2/\text{V-s}$ D) J/K
13. Which of the following is a piezoelectric material
 A) Lead B) Mica
 C) Iron D) Quartz
14. In a dielectric, the polarization is
 A) Linear function of applied field B) Square function of applied field
 C) Exponential functions of applied field D) Logarithmic function of applied field.
15. The energy due to dielectric loss appears as
 A) Light energy B) Sound energy
 C) Heat energy D) Electromagnetic energy
16. The unit of dipole moment/unit volume is
 A) Coulomb/meter B) Coulomb/meter²
 C) Coulomb/meter³ D) Coulomb
17. When a current is passed through the junction of two different metals, heat is absorbed or liberated depending on the direction of the current. What is this effect as?
 A) Seebeck B) Hall effect
 C) Thomson effect D) Peltier effect
18. In a thermocouple, the temperature that does not depend on the temperature of the cold junction is called
 A) Neutral temperature B) Temperature of inversion
 C) Absolute temperature D) Low temperature
19. The area enclosed within the hysteresis curve represents
 A) Degree of magnetization. B) Energy loss/unit volume/cycle.
 C) Number of domains/unit volume. D) Magnetic flux linkage/unit area.
20. The net magnetic moment in the paramagnetic substance in the absence of the external magnetic field is
 A) Infinite B) One
 C) Zero D) More than two

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the resonance and sharpness of resonance.	4	L*1	1	1,2
	b) What is force constant? Obtain an expression for a force constant when two springs are connected in series as well parallel.	8	L2	1	1,2
	c) A block whose mass m is 680 gm is fastened to a string whose spring constant k is 65 N/m. The block is pulled a distance x = 11 cm from its equilibrium position at x = 0 on a frictionless surface and released from rest at t = 0. What is the angular frequency, frequency and period of the resulting motion?	4	L3	1	1,2
2.	a) What is Poisson's ratio? Explain it.	4	L2	2	1,2
	b) Define the terms Y (Young's Modulus), K (bulk modulus), n (rigidity modulus). Give the relation connecting quantities.	8	L2	2	1,2
	c) A mass of 0.5 Kg hangs from a spring. If the mass is pulled downward and let go it executes SHM. Calculate the period if the same spring is stretched 16 cm by 0.4 kg mass.	4	L3	1	1,2

PH1007-1

SEE - May 2024

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| 3. | a) | Explain the stress and strain diagram. | 4 | L2 | 2 | 1,2 |
| | b) | Obtain the expression for the depression at the free end of a single cantilever and hence write an expression for Young's modulus. | 8 | L2 | 2 | 1,2 |
| | c) | What is the force required to stretch steel wire to double the length when it when it's area of cross section is 1 cm^2 ? Given Young's modulus 7×10^{10} Newton per meter square. | 4 | L3 | 2 | 1,2 |

Unit – II

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| 4. | a) | Explain the terms with neat diagram:
i) Direct & indirect band gap,
ii) P-N junction. | 4 | L2 | 3 | 1,2 |
| | b) | Obtain an expression for the conductivity of an extrinsic semiconductor. | 8 | L2 | 3 | 1,2 |
| | c) | An electric field of 100 Vm^{-1} is applied to a sample of n-type semiconductor having a Hall coefficient $-0.0125 \text{ m}^3\text{C}^{-1}$. Determine the current density in the sample if the electron mobility is $0.36 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$. | 4 | L3 | 3 | 1,2 |
| 5. | a) | Explain the any two types of polarization mechanism in dielectric. | 4 | L2 | 4 | 1,2 |
| | b) | What is Hall effect? Derive the expression for Hall coefficient and carrier concentration. | 8 | L2 | 3 | 1,2 |
| | c) | Sample of silicon is doped with 10^{22} phosphorus atoms m^{-3} . What would you expect to measure for its resistivity? What Hall voltage would you expect in a sample $100 \mu\text{m}$ thick when a current of 1 mA is passed perpendicular to a magnetic field of 0.1T .
[Given: $\mu_e = 0.07 \text{ m}^2\text{V}^{-1}\text{s}^{-1}$]. | 4 | L3 | 3 | 1,2 |
| 6. | a) | Write a note on:
(i) Ferroelectric materials,
(ii) Piezoelectric materials. | 4 | L1 | 4 | 1,2 |
| | b) | Derive an expression for one dimensional internal field developed in solid when an electric field is applied it. | 9 | L2 | 4 | 1,2 |
| | c) | Find the polarization produced in a dielectric medium of relative permittivity 15 in presence of an electric field of 500 V/m . | 3 | L3 | 4 | 1,2 |

Unit – III

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| 7. | a) | Explain the laws of thermoelectrics. | 4 | L2 | 5 | 1,2 |
| | b) | Explain the construction and working of a thermoelectric cooler. | 8 | L2 | 5 | 1,2 |
| | c) | The thermo e.m.f. of a Cu-Fe thermocouple is $2160 \mu\text{V}$ when the cold junction is at 0°C and the hot junction at 250°C . calculate the constant a and b if the neutral temperature is 330°C . | 4 | L3 | 5 | 1,2 |
| 8. | a) | Distinguish between soft and hard magnetic materials. | 4 | L1 | 5 | 1,2 |
| | b) | Explain the origin of magnetic moment, Bohr magnetron-electron, spin angular momentum of the electron with neat diagram | 8 | L2 | 5 | 1,2 |
| | c) | A silicon material is subjected to a magnetic field of strength 1000 A/m . if the magnetic susceptibility of silicon is -0.3×10^{-5} , calculate its magnetization. Also, evaluate the magnetic flux density of the field inside the materials. | 4 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
